Lab1 - Properties of LDRs

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1. Introduction

This laboratory investigation systematically characterizes the optoelectronic properties of light-dependent resistors (LDRs) through four complementary experimental approaches. The first two experiments revealed the basic photoelectric characteristics of the LDR, while the latter two experiments investigated its performance and potential limitations in practical applications. Together, these experiments provide a comprehensive understanding of the LDR's behavior and its potential for integration into electronic systems.

1.1 [Experiment 1] Investigating the "Light Intensity - Resistance" Changes of LDR under Different Light Conditions with Ground and Power Supply Connections

This experiment investigates the relationship between light intensity and electrical response in LDR circuits, by constructing two distinct configurations – a ground-connected LDR voltage divider and a supply-connected fixed resistor voltage divider.
When illuminated, the LDR's resistance decreases, leading to higher voltage across the fixed resistor in the upper-branch configuration and lower voltage in the lower-branch configuration.

1.2 [Experiment 2] Plotting the "Light Intensity - Resistance" Changes of LDR under Different Light Intensities

This experiment quantitatively measures the resistance changes of an LDR under controlled light intensity variations. Using a calibrated light source and a multimeter, the LDR's resistance ranges from complete darkness to bright light. The data was plotted to visualize a non-linear relationship between light intensity and resistance. The results reveal that the resistance decreases exponentially with increasing light intensity, suggesting a logarithmic dependence on illumination.

1.3 [Experiment 3] Comparison of the Change Range When LDR is Connected in Series with Different Voltage Divider Resistors

This experiment investigates that when the LDR is connected in series with a 1kΩ resistor, the change range from weak light to strong light is the largest. Larger resistors reduce the dynamic range, thus limiting the resolution of the circuit.

1.4 [Experiment 4] Investigating the Resistance Difference Between Two LDRs Under Identical Light Conditions

This experiment compares the resistance values of two different LDRs under same lighting conditions. Measurements revealed a resistance difference exceeding $200\ (\pm 5\ \Omega)$.

2. Theory and Technical Background

LDRs' register changing theory originally comes from semiconductor material. Those chemical material such as cadmium sulfide (CDs) offers election-hole pairs under light. Finally, the increasement of carrier concentration leads to higher electrical conductivity. This property turns out to be a negative correlation between light and resistance. The relationship between them is not linear. Usually satisfies the formula [1]. That is, a photoresistor is a semiconductor element based on the internal photoelectric effect, and its resistance value depends on the incident light intensity. As the incident light intensity increases, the resistance value of the photoresistor decreases, and as the incident light decreases, the resistance value of the photoresistor increases. Additionally, dark and bright light turns to totally different trends, with high resistance (up to 1.5 MΩ) in dark conditions opposing to a much lower resistive (small as 1KΩ or less).

$$R \propto E^{-\gamma}, \textit{where } \gamma \textit{ is material constant.} \qquad \textbf{[1]}$$

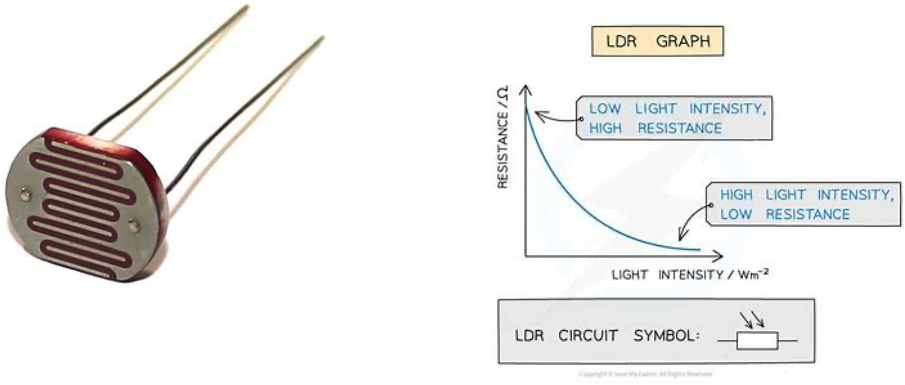


Figure 1: Photoresistors and their "Light Intensity-Resistance" curves

3. Methods:

Symbol	Description
ADC	Analog-to-Digital Conversion
IDE	Integrated Development Environment
LDR	Light-Dependent Resistor

Key words：Sixth-order polynomial fitting，*ADC*，Control variable

3.1 [Experiment 1] Investigating the "Resistance – Light Intensity" Changes of LDR under Different Light Conditions with Ground and Power Supply Connections

This experiment can be further divided into two sub-experiments: *LDR1 & LDR2* both connected to ground, and *LDR1 & LDR2* both connected to the power supply. (Fixed resistor resistance value: 1 kΩ)

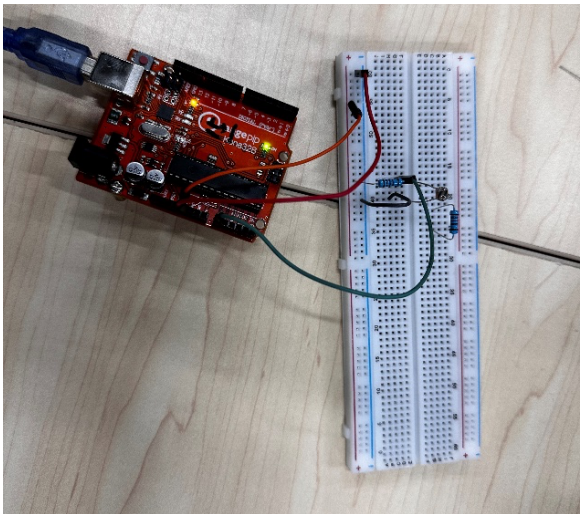


Figure 2: Grounding

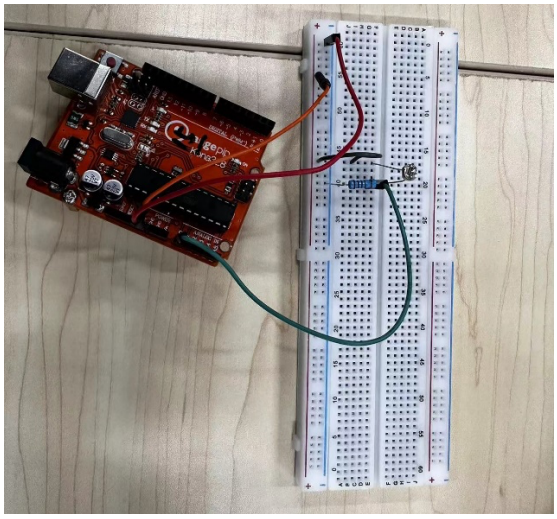


Figure 3: Power supply

The program is written using **Arduino IDE** to obtain the voltage values from the analog pins *A0* and *A1*, and convert them to digital values through *ADC*. The principle is as follows：

$$Voltage = \frac{V_{A_0}}{1023} \times \mathrm{V_{ref}} \quad \textbf{[2]}$$

$$V_{A_0} = V_{cc} \times \frac{R_L}{R_F + R_L} \quad \textbf{[3]}$$

$$R_L = R_F \times \frac{V_{ref} - V_{A_0}}{V_{A_0}} \quad \textbf{[4]}$$

Finally, get the corresponding *sensorValue1* and *sensorValue2* from the IDE's **Serial Monitor**. Then, using a 6th-order polynomial fitting, 300 points are inserted between the two data points to obtain a smooth curve. The methods of drawing of the image in the following experiments are the same.

3.2 [Experiment 2] Plotting the "Resistance – Light Intensity" Changes of LDR under Different Light Intensities

This experiment can be further turned into a sub-experiments: *LDR1* connected to the ground.

（Fixed resistor resistance value: 1 kΩ）

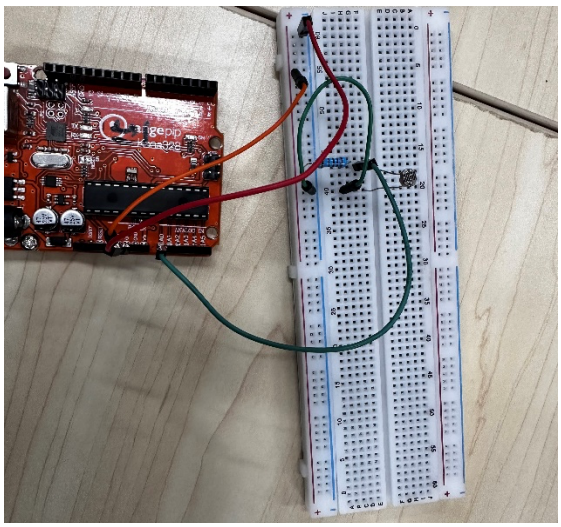


Figure 4: Grounding

The program is written using **Arduino IDE** to obtain the voltage value from analog pin - A0 and convert it to a digital value through *ADC*. The corresponding sensor Value is obtained through the **Serial Monitor** in the IDE.

3.3 [Experiment 3] Comparison of the Change Range When LDR is Connected in Series with Different Voltage Divider Resistors

- **Voltage divider resistance of 270Ω**

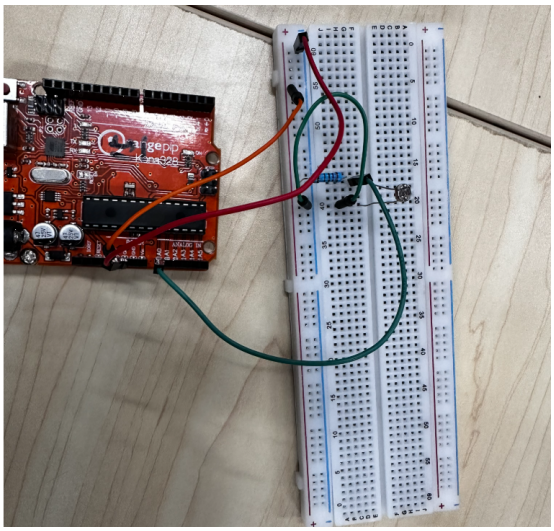


Figure 5: 270Ω Grounding

- **Voltage divider resistance of 1kΩ**

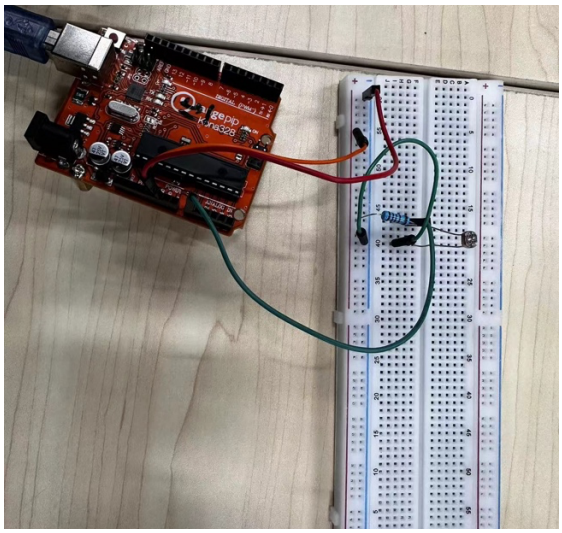


Figure 6: 1kΩ Grounding

- **Voltage divider resistance of 10kΩ：**

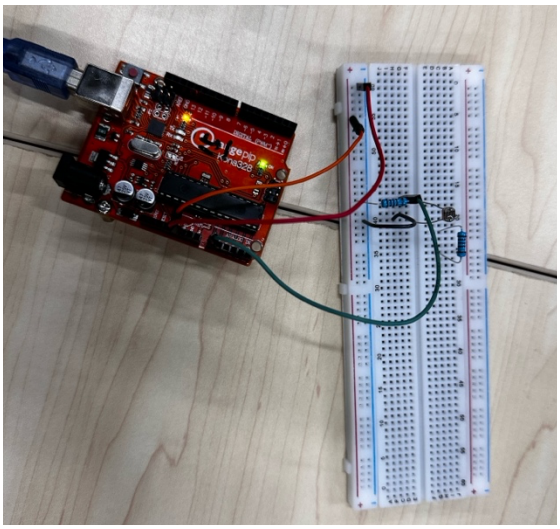


Figure 7: 10kΩ Grounding

After obtaining the value of the voltage on the analog pin - A0 after *ADC*, the polar differences are calculated and compared.

3.4 [Experiment 4] Investigating the Resistance Difference Between Two LDRs Under Identical Light Conditions

This experiment can be further divided into two sub-experiments: “ LDR1 & LDR2 both connected to ground ”, and “ LDR1 & LDR2 both connected to the power supply ”. *LDR1* and *LDR2* are placed in separate circuits with a controlled variable, keeping the voltage and the voltage divider resistors consistent.

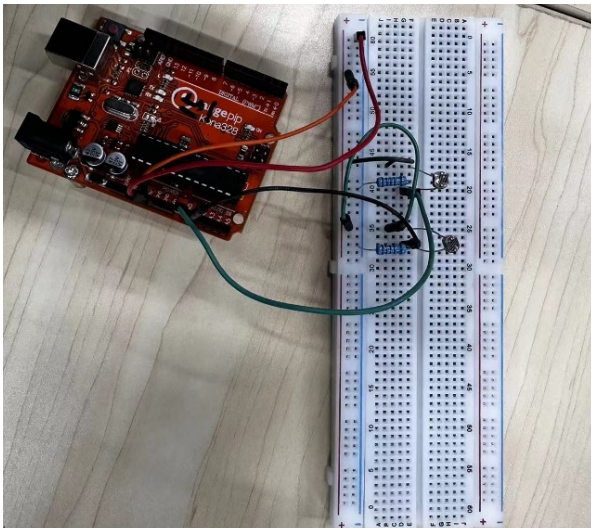


Figure 8: Both power supply

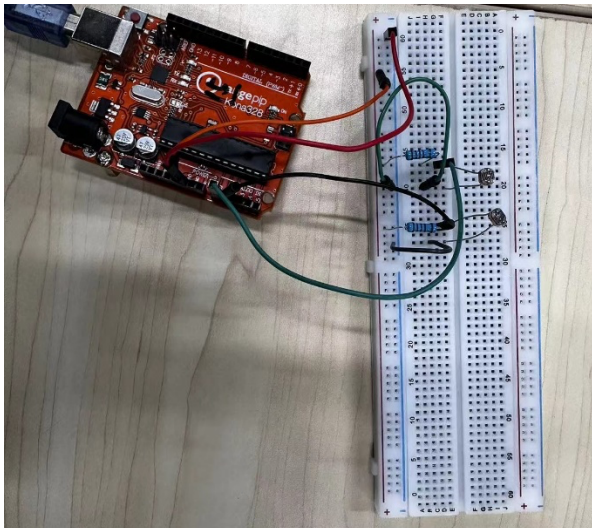


Figure 9: Both grounding

4. Results

4.1 Experiment 1

With fixed resistor resistance value 1kΩ, the corresponding *sensorValue* is obtained from the IDE's **Serial Monitor** as follows.

When LDR is grounded, the light goes from strong to weak:

GND_LDR_1:
[333, 256, 240, 246, 241, 225, 219, 225, 244, 240, 240, 307, 378, 450, 475, 508, 544, 584, 627, 660, 687, 712, 754, 773, 802, 817, 878, 1020, 1017, 1016, 1022, 1023, 1023, 1023]

GND_LDR_2: [153, 140, 133, 127, 122, 120, 119, 122, 124, 124, 129, 144, 174, 195, 221, 261, 303, 312, 343, 379, 397, 433, 464, 493, 537, 572, 592, 607, 593, 600, 626, 636, 631, 637, 654, 686, 705, 733, 781, 825, 844, 857, 858, 851, 850, 854, 1023, 1023, 1023, 1023, 1023, 1023, 1023]

When the LDR is connected to the power, the light goes from weak to strong:

VCC_LDR_1:
[141, 140, 137, 228, 434, 431, 406, 392, 394, 542, 543, 531, 521, 513, 584, 579, 591, 594, 659, 659, 653, 650, 649, 668, 680, 702, 758, 883, 822, 930, 949, 952, 960, 994, 1023, 1020, 1023, 1016, 1009, 1015]

VCC_LDR_2:
[172, 178, 181, 184, 277, 591, 585, 559, 550, 552, 757, 762, 768, 853, 853, 845, 846, 902, 897, 892, 885, 909, 911, 913, 914, 915, 913, 913, 916, 922, 956, 972, 982, 989, 1005, 1007, 1023, 1023, 1023, 1023]

The plotted curves are as follows:

GND_LDR:

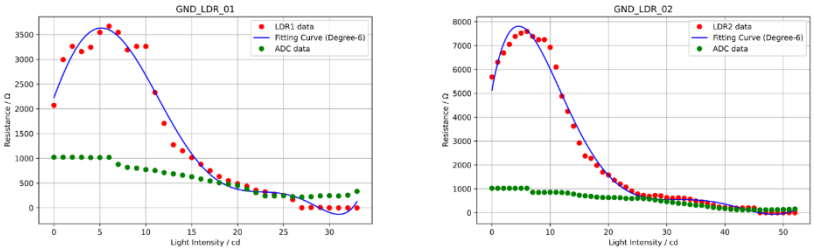


Figure 10: LDR1 & LDR2 Resistance - Light Intensity Curve at Ground Level

VCC_LDR:

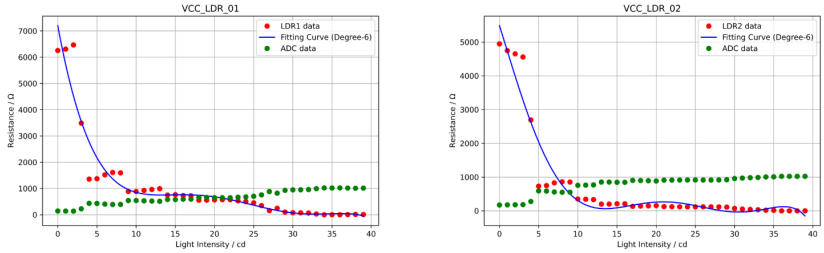


Figure 11: LDR1 & LDR2 Resistance - Light Intensity Curve at Power Supply

Through experimental data and images, we demonstrate that in both cases of LDR connected to power and ground. When connected to ground, the voltage drop across the light-dependent resistor (LDR) is measured. As the light intensity increases, the voltage drop across the LDR decreases from weak light to strong light. When connected to the power supply, the voltage drop across the resistor is measured. As the light intensity increases, the voltage drop across the resistor increases from weak light to strong light.

4.2 Experiment 2

With fixed resistor resistance value 1kΩ and *LDR1*, the corresponding *sensorValue* is obtained through the IDE's **serial monitor** as follows:

```
GND_LDR_1:
[ 333, 256, 240, 246, 241, 225, 219, 225, 244, 240, 240, 307, 378, 450, 475, 508, 544, 584, 627, 660,
687, 712, 754, 773, 802, 817, 878, 1020, 1017, 1016, 1022, 1023, 1023, 1023 ]
```

The plotted curves are as follows:

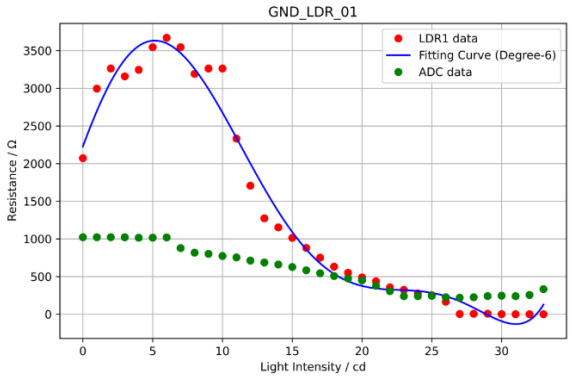


Figure 12: LDR1 Resistance - Light Intensity Curve at Ground Level

It can be illustrated from the experimental data and images that the resistance value of the *LDR* decreases with increasing light, it should be noted that the experimental image does not strictly follow the light from if to strong, the image initially appears to be rising.
The reason for the rise will be explained later.

4.3 Experiment 3

- **Measured data and curve when the voltage divider resistor is 270Ω shows as follow:**

VCC_LDR_1: [1023, 1023, 1023, 1023, 1023, 1023, 1023, 1023, 1023, 1023, 1023, 892, 880, 887, 788, 796, 716, 723, 731, 717, 680, 678, 600, 656, 619, 560, 541, 508, 480, 456, 437, 399, 432, 492, 524, 831, 1023]

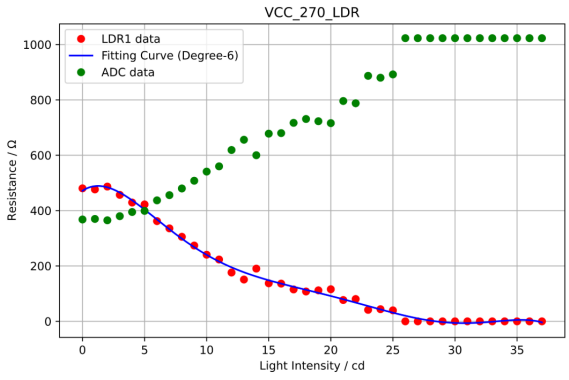


Figure 13: LDR1 with 270Ω Resistance- Light Intensity Curve at Ground Level

- **Measured data and curve when the voltage divider resistor is 1kΩ shows as follows:**

VCC_LDR_1:
[141, 140, 137, 228, 434, 431, 406, 392, 394, 542, 543, 531, 521, 513, 584, 579, 591, 594, 659, 659, 653, 650, 649, 668, 680, 702, 758, 883, 822, 930, 949, 952, 960, 994, 1023, 1020, 1023, 1016, 1009, 1015]

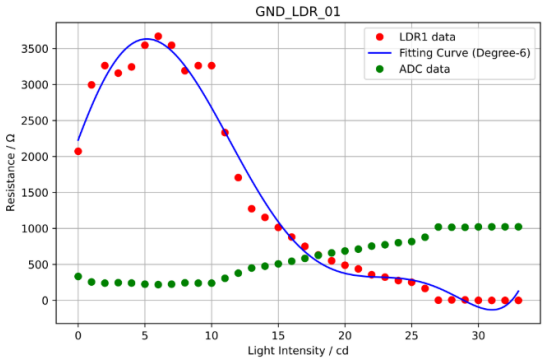


Figure 14: LDR1 with 1kΩ Resistance- Light Intensity Curve at Ground Level

- **Measured data and curve when the voltage divider resistor is 10kΩ shows as follow:**

VCC_LDR_1: [606, 605, 618, 651, 212, 206, 206, 206, 71, 68, 66, 63, 60, 61, 42, 43, 42, 33, 33, 33, 33, 31, 31, 31, 31, 32, 32, 38, 48]

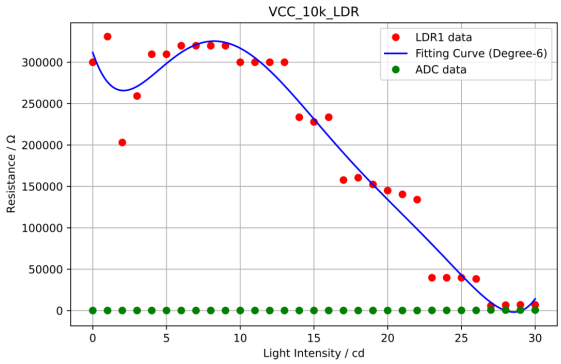


Figure 15: LDR1 with 10kΩ Resistance- Light Intensity Curve at Ground Level

Table 1: The polar differences of different resistance

Resistance	Polar difference
270 Ω	**624**
1 kΩ	**690**
10 kΩ	**558**

By calculating the polarity difference, we find that 1kΩ has the largest polarity difference. That is to say when 1kΩ is used as a voltage divider resistor, the change range from weak light to strong light is the largest.

4.4 Experiment 4

When the light is swept from *LDR1* to *LDR2*, the following data is obtained:

VCC_LDR_1:
[664, 753, 840, 889, 925, 973, 1015, 1023, 1023, 1023, 1023, 1023, 1023, 1023, 997, 919, 780, 693, 655, 631, 620, 617, 623, 602, 593, 598]

VCC_LDR_2:
[458, 484, 530, 557, 585, 619, 647, 730, 783, 830, 881, 941, 987, 1020, 1019, 987, 942, 822, 723, 618, 528, 511, 512, 519, 433, 429, 433]

When the light is swept from LDR2 to LDR1, the following data is obtained:

GND_LDR_1: [878, 784, 756, 697, 664, 622, 548, 473, 418, 410, 430, 476, 521, 590, 727, 802, 828, 843, 844, 842, 837, 863]

GND_LDR_2: [1023, 975, 958, 926, 900, 870, 820, 739, 649, 610, 605, 615, 641, 695, 832, 964, 988, 1003, 1004, 1004, 998, 1023]

The experimentally obtained curves are as follows:

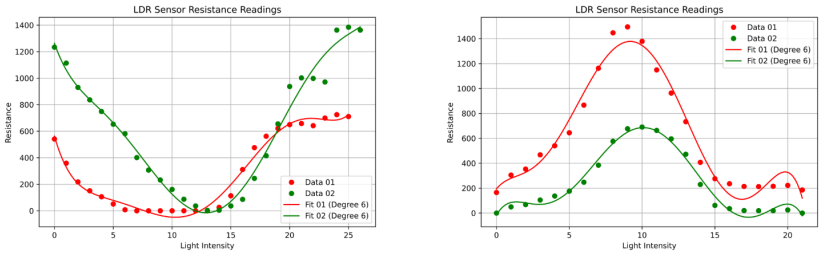


Figure 16: Light swept from LDR2 to LDR1 & from LDR1 to LDR2

Through the data and curves we can see that in the same lighting environment, the resistance values of *LDR1* and *LDR2* maintain a relatively large difference, a value that can exceed a maximum of 200Ω under polarized light conditions.

5. Conclusion

- **Conclusion and Summary:**

This experiment consists of four sub-experiments:

1) It verifies the characteristic of the LDR, where its resistance decreases as the light intensity increases.

2) It’s determined that, for these two LDRs, using a 1kΩ series resistor in the voltage divider results in the maximum range of resistance variation from darkness to bright light. This finding will assist in selecting an appropriate voltage divider resistor in subsequent experiments to enhance the sensitivity of the LDR's response.

3) It’s observed that the resistance difference between two LDRs under different lighting conditions in the same light environment is significant, providing a reliable basis for using the resistance difference of the two LDRs as a judgment criterion in future experiments.

- **Limitations of the experiment:**

1) The experimental environment could not fully eliminate the influence of the laboratory lighting, which led to the observed curve exhibiting an initial increase followed by a decrease.

2) Achieving a smooth variation in light intensity proved challenging, resulting in abrupt changes in the data and the curve.

6. Appendix

- **The code used for plotting is as follows:**

```python
import numpy as np
import mlx.core as mx
import matplotlib.pyplot as plt

R_Fixed = 1000.0
Vref = 5.0

adc_values_01 = np.array([
    172, 178, 181, 184, 277, 591, 585, 559, 550, 552, 757, 762, 768, 853,
    853, 845, 846, 902, 897, 892, 885, 909, 911, 913, 914, 915, 913, 913,
    916, 922, 956, 972, 982, 989, 1005, 1007, 1023, 1023, 1023, 1023
])

Vout = adc_values_01 / 1023.0 * Vref

R_LDR = R_Fixed * ((Vref - Vout) / Vout)

x_values = mx.arange(stop=len(adc_values_01))

poly_degree = 6
poly_coeffs_01 = np.polyfit(x_values, R_LDR, poly_degree)
poly_func_01 = np.poly1d(poly_coeffs_01)

x_dense = mx.linspace(start=x_values.min(), stop=x_values.max(), num=300)
y_dense = poly_func_01(x_dense)

plt.figure(figsize=(8, 5))
plt.plot(x_values, R_LDR, 'ro', label="LDR2 data")
plt.plot(x_dense, y_dense, 'b-', label=f"Fitting Curve (Degree-{poly_degree})")
plt.plot(x_values, adc_values_01, 'go', label="ADC data")

plt.xlabel("Light Intensity / cd")
plt.ylabel(r"Resistance / $\Omega$")
plt.title("VCC_LDR_02")
plt.grid(True)
plt.legend()

plt.savefig("./Photos/Chart/VCC_LDR_02.png", dpi=1200)

plt.show()

```

- **The code used to get the ADC value is as follows:**

```arduino
void setup() {
    // put your setup code here, to run once:
    Serial.begin(9600);
}

void loop() {
    // put your main code here, to run repeatedly:
    int sensorValue_A0 = analogRead(A0);

    Serial.print("Sensor Value of A0 = ");
    Serial.println(sensorValue_A0);

    delay(1);
    delay(1000);
}

```